

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M.Tech. (CE)

December 2012

CE701.01 Operating System Design & Concepts

Date: 18.12.2012, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) How IPC will work using files? Is this safe method to perform IPC? Justify your answer with proper example. [04]
- (b) Define the steps to build a kernel under BSD based systems? [03]
- (c) What happen when a thread request a block of data from the file system? [02]
- (d) Write your own function called **dup2()**. [02]
- Q - 2 (a) Suppose a process does not wish to block until its children terminate. How can it ensure that child processes are cleaned up when they terminate? Explain in detail. [04]
- (b) How Hybrid kernels are Microkernels with aspects from Monolithic kernels? [04]

OR

- (b) How does the dispatcher get control back during context switch? [04]
- (c) What is LWP (Light weight process)? Explain thread scheduling with diagram in detail [04]

OR

- (c) What are the basic scheduling scopes for threads? Explain all of them in detail. [04]

- Q - 3 Answer the following questions. (Any Four) [12]

- (1) The Linux kernel does not allow paging out of kernel memory. What effect does this restriction have on the kernel's design? What are two advantages and two disadvantages of this design decision?
- (2) Explain how the process represented in system memory.
- (3) Differentiate followings:
 - (i) Synchronous and Asynchronous process execution
 - (ii) **fork()** and **vfork()**
- (4) What is shared memory? How to make the shared memory segment available using proper example.
- (5) Why does each LWP need a separate kernel stack? Can the system save resources by allocating a kernel stack only when an LWP makes a system call?

SECTION – II

Q-4 Answer the following questions.

- (a) Draw and explain in brief the architecture of UNIX system. [02]
- (b) What is the requirement of buffer cache in UNIX system? [02]
- (c) Write an algorithm for releasing a buffer. [02]
- (d) Explain the architecture of Google file system. [02]
- (e) Explain the advantage of Exokernel operating system over traditional operating system. [03]

Q-5 Answer the following questions.

- (a) Explain the scenarios for retrieval of a buffer in buffer cache. [04]
- (b) Explain race condition for unlink system call. [08]

OR

- (b) Write & Explain an algorithm for open system call in UNIX system. [08]

Q-6 Answer the following questions. (Attempt any three)

[12]

- (a) Explain STAT and FSTAT system call in brief.
- (b) Draw and explain the architecture of Google File System.
- (c) Write and explain close system call in detail.
- (d) Write a program that prints the owner, file type access permissions, and access, times of files supplied as parameters. If a file is a directory, the program should read the directory and print the above information for all files in the directory.
